```
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int request[50], n, head, disk;
    int i, j, temp, movement = 0;

    printf("Enter number of requests: ");
    scanf("%d", &n);

    printf("Enter request queue:\n");

    for(i = 0; i < n; i++)
        scanf("%d", &request[i]);

    printf("Enter initial head position: ");
    scanf("%d", &head);

    printf("Enter disk size: ");
    scanf("%d", &disk);

    for(i = 0; i < n - 1; i++)
    {
        for(j = i + 1; j < n; j++)
        {
            if(request[i] > request[j])
            {
                temp = request[i];
                request[i] = request[j];
                request[j] = temp;
```

```
        }
    }
}

printf("\nSeek Sequence: %d", head);

/* Move right */
for(i = 0; i < n; i++)
{
    if(request[i] >= head)
    {
        movement += abs(head - request[i]);
        head = request[i];
        printf(" -> %d", head);
    }
}

/* Move to end */
movement += (disk - 1) - head;
head = disk - 1;
printf(" -> %d", head);

/* Jump to beginning */
movement += disk - 1;
head = 0;
printf(" -> %d", head);

/* Service remaining requests */
for(i = 0; i < n; i++)
{
    if(request[i] < head)
```

```
    {
        movement += abs(head - request[i]);
        head = request[i];
        printf(" -> %d", head);
    }
  }

  printf("\nTotal Head Movement = %d cylinders\n", movement);

  return 0;
}
```

OUTPUT:

Enter number of requests: 8

Enter request queue:

98 183 37 122 14 124 65 67

Enter initial head position: 53

Enter disk size: 200

Seek Sequence: 53 -> 65 -> 67 -> 98 -> 122 -> 124 -> 183 -> 199 -> 0 -> 14 -> 37

Total Head Movement = 382 cylinders